

Detailed Derivation of the Hubbard Contribution to $\Delta\mathbf{M}$ and Its Analogy with $\Delta\mathbf{M}_{\text{bare}}$

1 Goal

The purpose of this note is to derive step by step the contribution to the orbital magnetization associated with a Hubbard potential, starting from the same GIPAW-like expression used for the bare non-local contribution. The goal is to show that the final Hubbard term has exactly the same structure as $\Delta\mathbf{M}_{\text{bare}}$, upon replacing non-local pseudopotential projectors with Hubbard projectors.

Throughout the derivation we assume the norm-conserving case, so that the overlap operator is the identity:

$$S = \mathbb{I}. \quad (1)$$

Therefore no ultrasoft overlap corrections appear.

2 Starting point

We start from the standard bare contribution written in operator form:

$$\Delta\mathbf{M} = \frac{\alpha}{2} \sum_I \left\langle (\mathbf{R}_I - \mathbf{r}) \times \frac{1}{i} [\mathbf{R}_I - \mathbf{r}, \hat{V}_I] \right\rangle. \quad (2)$$

Here:

- I labels the atomic site,
- $\mathbf{R}_I$ is the position of site I ,
- $\hat{V}_I$ is an operator localized on site I .

For convenience, define

$$\boldsymbol{\rho}_I = \mathbf{r} - \mathbf{R}_I. \quad (3)$$

Then

$$\mathbf{R}_I - \mathbf{r} = -\boldsymbol{\rho}_I, \quad (4)$$

and

$$[\mathbf{R}_I - \mathbf{r}, \hat{V}_I] = [-\boldsymbol{\rho}_I, \hat{V}_I] = -[\boldsymbol{\rho}_I, \hat{V}_I]. \quad (5)$$

Therefore Eq. (2) becomes

$$\Delta\mathbf{M} = \frac{\alpha}{2} \sum_I \left\langle \boldsymbol{\rho}_I \times \frac{1}{i} [\boldsymbol{\rho}_I, \hat{V}_I] \right\rangle. \quad (6)$$

3 Expansion of the commutator

We expand the commutator:

$$[\boldsymbol{\rho}_I, \hat{V}_I] = \boldsymbol{\rho}_I \hat{V}_I - \hat{V}_I \boldsymbol{\rho}_I. \quad (7)$$

Substituting into Eq. (6), we obtain

$$\Delta \mathbf{M} = \frac{\alpha}{2} \sum_I \left\langle \boldsymbol{\rho}_I \times \frac{1}{i} (\boldsymbol{\rho}_I \hat{V}_I - \hat{V}_I \boldsymbol{\rho}_I) \right\rangle. \quad (8)$$

Using linearity of the cross product,

$$\boldsymbol{\rho}_I \times (\boldsymbol{\rho}_I \hat{V}_I - \hat{V}_I \boldsymbol{\rho}_I) = \boldsymbol{\rho}_I \times (\boldsymbol{\rho}_I \hat{V}_I) - \boldsymbol{\rho}_I \times (\hat{V}_I \boldsymbol{\rho}_I). \quad (9)$$

Let us analyze the first term. In components,

$$[\boldsymbol{\rho}_I \times (\boldsymbol{\rho}_I \hat{V}_I)]_\gamma = \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \rho_{I,\mu} \rho_{I,\nu} \hat{V}_I. \quad (10)$$

Since $\rho_{I,\mu} \rho_{I,\nu}$ is symmetric under $\mu \leftrightarrow \nu$, while $\epsilon_{\gamma\mu\nu}$ is antisymmetric, this term vanishes:

$$\boldsymbol{\rho}_I \times (\boldsymbol{\rho}_I \hat{V}_I) = 0. \quad (11)$$

Therefore,

$$\boldsymbol{\rho}_I \times (\boldsymbol{\rho}_I \hat{V}_I - \hat{V}_I \boldsymbol{\rho}_I) = -\boldsymbol{\rho}_I \times (\hat{V}_I \boldsymbol{\rho}_I). \quad (12)$$

Hence Eq. (8) becomes

$$\Delta \mathbf{M} = -\frac{\alpha}{2} \sum_I \left\langle \boldsymbol{\rho}_I \times \frac{1}{i} \hat{V}_I \boldsymbol{\rho}_I \right\rangle. \quad (13)$$

Equivalently, since $1/i = -i$,

$$\Delta \mathbf{M} = \frac{i\alpha}{2} \sum_I \left\langle \boldsymbol{\rho}_I \times \hat{V}_I \boldsymbol{\rho}_I \right\rangle. \quad (14)$$

4 Recovery of the bare non-local formula

Consider first a separable non-local pseudopotential:

$$\hat{V}_I^{\text{NL}} = \sum_{ij} |\beta_i^I\rangle D_{ij}^I \langle \beta_j^I|. \quad (15)$$

Substituting Eq. (15) into Eq. (13), we obtain

$$\Delta \mathbf{M}_{\text{bare}} = -\frac{\alpha}{2} \sum_I \left\langle \boldsymbol{\rho}_I \times \frac{1}{i} \sum_{ij} |\beta_i^I\rangle D_{ij}^I \langle \beta_j^I| \boldsymbol{\rho}_I \right\rangle. \quad (16)$$

By linearity,

$$\Delta \mathbf{M}_{\text{bare}} = -\frac{\alpha}{2} \sum_I \sum_{ij} \left\langle \boldsymbol{\rho}_I \times \frac{1}{i} |\beta_i^I\rangle D_{ij}^I \langle \beta_j^I| \boldsymbol{\rho}_I \right\rangle. \quad (17)$$

Evaluating the expectation value over occupied Bloch states gives

$$\Delta \mathbf{M}_{\text{bare}} = -\frac{\alpha}{2} \sum_{\mathbf{n}\mathbf{k}}^{\text{occ}} f_{\mathbf{n}\mathbf{k}} \sum_I \sum_{ij} \left\langle u_{\mathbf{n}\mathbf{k}} \left| \boldsymbol{\rho}_I \times \frac{1}{i} |\beta_i^{\mathbf{k},I}\rangle D_{ij}^I \langle \beta_j^{\mathbf{k},I}| \boldsymbol{\rho}_I \right| u_{\mathbf{n}\mathbf{k}} \right\rangle. \quad (18)$$

Using

$$(\mathbf{A} \times \mathbf{B})_\gamma = \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} A_\mu B_\nu, \quad (19)$$

we obtain

$$\Delta M_{\text{bare},\gamma} = -\frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_I \sum_{ij} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \frac{1}{i} \langle u_{n\mathbf{k}} | \rho_{I,\mu} | \beta_i^{\mathbf{k},I} \rangle D_{ij}^I \langle \beta_j^{\mathbf{k},I} | \rho_{I,\nu} | u_{n\mathbf{k}} \rangle. \quad (20)$$

More precisely, for a projector localized on atom I , we use the no-phase KB projector, i.e. the projector in which the global $e^{-i\mathbf{k} \cdot \mathbf{R}_I}$ phase associated with the atomic center has been removed. This convention is discussed in Appendix A. With this choice, the dependence on the local coordinate

$$\boldsymbol{\rho}_I = \mathbf{r} - \mathbf{R}_I \quad (21)$$

is made explicit by writing

$$\tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) = e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{R}_I)} \beta_i^I(\mathbf{r}), \quad (22)$$

where $\beta_i^I(\mathbf{r})$ denotes the localized projector already centered on atom I .

It follows that

$$\partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) = -i(r_\mu - R_{I,\mu}) \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) = -i\rho_{I,\mu} \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}). \quad (23)$$

Hence,

$$\rho_{I,\mu} \tilde{\beta}_i^{\mathbf{k},I} = i \partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I}. \quad (24)$$

This gives

$$\langle u_{n\mathbf{k}} | \rho_{I,\mu} | \tilde{\beta}_i^{\mathbf{k},I} \rangle = i \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I} \rangle, \quad (25)$$

and, by Hermitian conjugation,

$$\langle \tilde{\beta}_j^{\mathbf{k},I} | \rho_{I,\nu} | u_{n\mathbf{k}} \rangle = -i \langle \partial_{k_\nu} \tilde{\beta}_j^{\mathbf{k},I} | u_{n\mathbf{k}} \rangle. \quad (26)$$

Substituting into Eq. (20),

$$\begin{aligned} \Delta M_{\text{bare},\gamma} &= -\frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_I \sum_{ij} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \frac{1}{i} \left(i \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I} \rangle \right) D_{ij}^I \left(-i \langle \partial_{k_\nu} \tilde{\beta}_j^{\mathbf{k},I} | u_{n\mathbf{k}} \rangle \right) \\ &= \frac{i\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_I \sum_{ij} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I} \rangle D_{ij}^I \langle \partial_{k_\nu} \tilde{\beta}_j^{\mathbf{k},I} | u_{n\mathbf{k}} \rangle. \end{aligned} \quad (27)$$

Since the contraction with $\epsilon_{\gamma\mu\nu}$ selects the antisymmetric part in the Cartesian indices, the result can be written in manifestly real form as

$$\Delta M_{\text{bare},\gamma} = -\frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_I \sum_{ij} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \text{Im} \left[\langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I} \rangle D_{ij}^I \langle \partial_{k_\nu} \tilde{\beta}_j^{\mathbf{k},I} | u_{n\mathbf{k}} \rangle \right]. \quad (28)$$

Note. In the implementation the full double sum over (μ, ν) is not performed explicitly. For a given magnetization component γ , only one cyclic pair (μ, ν) is evaluated. For instance,

$$(\gamma, \mu, \nu) = (1, 2, 3), (2, 3, 1), (3, 1, 2),$$

so that $\epsilon_{\gamma\mu\nu} = +1$.

Let

$$Z_{\mu\nu} = \frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_I \sum_{ij} \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I} \rangle D_{ij}^I \langle \partial_{k_\nu} \tilde{\beta}_j^{\mathbf{k},I} | u_{n\mathbf{k}} \rangle. \quad (29)$$

Using the Hermiticity of D_{ij}^I , one has

$$Z_{\nu\mu} = Z_{\mu\nu}^*. \quad (30)$$

Therefore the two antisymmetric terms selected by $\epsilon_{\gamma\mu\nu}$ give

$$\begin{aligned} \epsilon_{\gamma\mu\nu} \text{Im } Z_{\mu\nu} + \epsilon_{\gamma\nu\mu} \text{Im } Z_{\nu\mu} &= \text{Im } Z_{\mu\nu} - \text{Im } Z_{\nu\mu} \\ &= \text{Im } Z_{\mu\nu} - \text{Im } Z_{\mu\nu}^* \\ &= 2 \text{Im } Z_{\mu\nu}. \end{aligned} \quad (31)$$

Thus, evaluating only one cyclic Cartesian pair requires multiplying the result by a factor of two. With the overall negative sign in Eq. (28), the implemented contribution is therefore proportional to

$$-2 \text{Im } Z_{\mu\nu}. \quad (32)$$

This corresponds to the code statement

```
delta_M_bare(kk) = delta_M_bare(kk) - 2.d0*imag(tmp)
```

where `tmp` is the numerical realization of $Z_{\mu\nu}$ for the cyclic pair associated with the component `kk`.

5 Hubbard potential in separable form

In the norm-conserving case, the Hubbard potential localized on site I can be written in separable form as

$$\hat{V}_I^{\text{Hub}} = \sum_{mm'} |\phi_{Im}^{\mathbf{k}}\rangle V_{mm'}^I \langle \phi_{Im'}^{\mathbf{k}}|. \quad (33)$$

Here:

- $|\phi_{Im}^{\mathbf{k}}\rangle$ and $\langle \phi_{Im'}^{\mathbf{k}}|$ denote the Bloch-summed Hubbard projectors associated with site I ;
- depending on the chosen Hubbard-projector scheme, these projectors may be atomic orbitals, orthogonalized atomic orbitals, or projectors read from the pseudopotential file;
- m, m' label the local Hubbard manifold;
- $V_{mm'}^I$ is the Hubbard potential matrix on site I .

6 Derivation of the Hubbard contribution

We now repeat the same derivation with $\hat{V}_I = \hat{V}_I^{\text{Hub}}$. Starting from Eq. (13),

$$\Delta \mathbf{M}_{\text{Hub}} = -\frac{\alpha}{2} \sum_I \left\langle \boldsymbol{\rho}_I \times \frac{1}{i} \hat{V}_I^{\text{Hub}} \boldsymbol{\rho}_I \right\rangle. \quad (34)$$

Substituting Eq. (33),

$$\Delta \mathbf{M}_{\text{Hub}} = -\frac{\alpha}{2} \sum_I \left\langle \boldsymbol{\rho}_I \times \frac{1}{i} \sum_{mm'} V_{mm'}^I \phi_{Im}^{\mathbf{k}} \phi_{Im'}^{\mathbf{k}} \boldsymbol{\rho}_I \right\rangle. \quad (35)$$

By linearity,

$$\Delta \mathbf{M}_{\text{Hub}} = -\frac{\alpha}{2} \sum_I \sum_{mm'} \left\langle \boldsymbol{\rho}_I \times \frac{1}{i} V_{mm'}^I \phi_{Im}^{\mathbf{k}} \phi_{Im'}^{\mathbf{k}} \boldsymbol{\rho}_I \right\rangle. \quad (36)$$

Evaluating over occupied Bloch states,

$$\Delta \mathbf{M}_{\text{Hub}} = -\frac{\alpha}{2} \sum_{n\mathbf{k}}^{\text{occ}} f_{n\mathbf{k}} \sum_I \sum_{mm'} \left\langle u_{n\mathbf{k}} \left| \boldsymbol{\rho}_I \times \frac{1}{i} V_{mm'}^I \phi_{Im}^{\mathbf{k}} \phi_{Im'}^{\mathbf{k}} \boldsymbol{\rho}_I \right| u_{n\mathbf{k}} \right\rangle. \quad (37)$$

Using Eq. (19),

$$\Delta M_{\text{Hub},\gamma} = -\frac{\alpha}{2} \sum_{n\mathbf{k}}^{\text{occ}} f_{n\mathbf{k}} \sum_I \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \frac{1}{i} \langle u_{n\mathbf{k}} | \rho_{I,\mu} | \phi_{Im}^{\mathbf{k}} \rangle V_{mm'}^I \langle \phi_{Im'}^{\mathbf{k}} | \rho_{I,\nu} | u_{n\mathbf{k}} \rangle. \quad (38)$$

In complete analogy with the KB projectors, the Hubbard projectors entering the k derivatives are understood to be no-phase projectors. We denote them by $\tilde{\phi}_{Im}^{\mathbf{k}}$. The tilde will be omitted in the final compact expressions only when no ambiguity is possible.

Now use the analogues of Eqs. (25) and (26) for the no-phase Hubbard projectors:

$$\langle u_{n\mathbf{k}} | \rho_{I,\mu} | \tilde{\phi}_{Im}^{\mathbf{k}} \rangle = i \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_{Im}^{\mathbf{k}} \rangle, \quad (39)$$

and

$$\langle \tilde{\phi}_{Im'}^{\mathbf{k}} | \rho_{I,\nu} | u_{n\mathbf{k}} \rangle = -i \langle \partial_{k_\nu} \tilde{\phi}_{Im'}^{\mathbf{k}} | u_{n\mathbf{k}} \rangle. \quad (40)$$

Substituting into Eq. (38),

$$\begin{aligned} \Delta M_{\text{Hub},\gamma} &= -\frac{\alpha}{2} \sum_{n\mathbf{k}}^{\text{occ}} f_{n\mathbf{k}} \sum_I \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \frac{1}{i} \left(i \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_{Im}^{\mathbf{k}} \rangle \right) V_{mm'}^I \left(-i \langle \partial_{k_\nu} \tilde{\phi}_{Im'}^{\mathbf{k}} | u_{n\mathbf{k}} \rangle \right) \\ &= \frac{i\alpha}{2} \sum_{n\mathbf{k}}^{\text{occ}} f_{n\mathbf{k}} \sum_I \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_{Im}^{\mathbf{k}} \rangle V_{mm'}^I \langle \partial_{k_\nu} \tilde{\phi}_{Im'}^{\mathbf{k}} | u_{n\mathbf{k}} \rangle. \end{aligned} \quad (41)$$

Again, since the contraction with $\epsilon_{\gamma\mu\nu}$ selects the antisymmetric part in the Cartesian indices, the result can be written in manifestly real form as

$$\Delta M_{\text{Hub},\gamma} = -\frac{\alpha}{2} \sum_{n\mathbf{k}}^{\text{occ}} f_{n\mathbf{k}} \sum_I \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \text{Im} \left[\langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_{Im}^{\mathbf{k}} \rangle V_{mm'}^I \langle \partial_{k_\nu} \tilde{\phi}_{Im'}^{\mathbf{k}} | u_{n\mathbf{k}} \rangle \right]. \quad (42)$$

In the present norm-conserving case, with $S = \mathbb{I}$, the matrix $V_{mm'}^I$ appearing in the separable Hubbard operator is identified with the on-site Hubbard potential matrix

$$V_{mm'}^I \equiv U_{mm'}^{I,\sigma} = U^I \left[\frac{\delta_{mm'}}{2} - n_{mm'}^{I,\sigma} \right],$$

with the spin index σ understood implicitly whenever not written explicitly.

The quantity

$$\langle \partial_{k_\nu} \tilde{\phi}_{Im'}^{\mathbf{k}} | u_{n\mathbf{k}} \rangle$$

is evaluated numerically by the routine `compute_projU`, reported below, which is the direct analogue of `compute_dbecp` for the Hubbard projectors (`wfcU` in QE). In the present implementation, the explicit stripping of the Bloch phase has not yet been introduced; however, tests indicate that this has only a minor impact on the numerical result.

```
!-----
! derivative of the wfcU (projU)
!-----
SUBROUTINE compute_projU
  USE cell_base, ONLY : tpiba
  USE uspp_init, ONLY : init_us_2
  USE uspp, ONLY : vkb
  USE ldaU, ONLY : wfcU
  implicit none
  integer :: ipol, sig
  integer :: iu
  real(dp) :: kq(3), xk_save(3)

  call deallocate_bec_type(becp)
  xk_save(:) = xk(:,ik)

  if (nkb == 0) return
  wfcU_save = wfcU

  do ipol = 1, 3
    projU(:, :, ipol) = (0.d0, 0.d0)
    do sig = -1, 1, 2
      xk(:, ik) = xk_save(:)
      xk(ipol, ik) = xk(ipol, ik) + sig * delta_k
      call orthoUwfc_k( ik, .FALSE. )
      call strip_kphase_wfcU( ik )
      call calbec( npw, wfcU, evc, auxU, nbnd )
      projU(:, :, ipol) = projU(:, :, ipol) + &
        0.5d0*sig/(delta_k*tpiba) * auxU(:, :)
    enddo
  enddo

  xk(:, ik) = xk_save(:)
  wfcU = wfcU_save
  CALL allocate_bec_type(nkb, nbnd, becp)
END SUBROUTINE compute_projU
```

The following routine removes the k -dependent Bloch phase from the Hubbard projectors before evaluating the non-local Hubbard contribution to the orbital magnetization.

```
SUBROUTINE strip_kphase_wfcU(ik)
  USE kinds, ONLY : DP
  USE constants, ONLY : tpi ! tpi = 2*pi in QE
  USE klist, ONLY : xk
```

```

USE ions_base, ONLY : nat, tau, ityp
USE ldaU,          ONLY : wfcU, nwfcU, Hubbard_l ! Hubbard_l(ntyp)
IMPLICIT NONE

INTEGER, INTENT(IN) :: ik
INTEGER :: na, nt, l, m, iu, nproj
REAL(DP) :: arg
COMPLEX(DP) :: phase

iu = 0

DO na = 1, nat
  nt = ityp(na)
  l = Hubbard_l(nt)

  ! Convenzione tipica: l < 0 => no Hubbard on this type
  IF (l >= 0) THEN
    nproj = 2*l + 1 ! collinear / NC

    arg = tpi * ( xk(1,ik)*tau(1,na) + xk(2,ik)*tau(2,na) + xk(3,ik)*tau(3,na) )
    phase = CMPLX( COS(arg), SIN(arg), KIND=DP ) ! exp(+i arg)

    DO m = 1, nproj
      iu = iu + 1
      wfcU(:,iu) = phase * wfcU(:,iu)
    END DO
  END IF
END DO

IF (iu /= nwfcU) THEN
  CALL errore('strip_kphase_wfcU', 'unexpected wfcU ordering: iu != nwfcU', 1)
END IF
END SUBROUTINE strip_kphase_wfcU

```

The final Hubbard contribution in Eq. (42) is evaluated by the routine reported below:

```

!-----
! GIPAW correction (delta_M_Hub)
!-----
SUBROUTINE calc_delta_M_hub
  USE ions_base, ONLY : nat, ntyp => nsp, ityp
  USE ldaU,       ONLY : Hubbard_l, is_hubbard, offsetU, nwfcU
  USE scf,        ONLY : v
  implicit none
  complex(dp) :: tmp, hub_product
  integer :: ibnd, na, nt, m1, m2, ldim, off

  if (nwfcU == 0) return

  tmp = (0.d0, 0.d0)

```

```

do nt = 1, ntyp
  if (.not. is_hubbard(nt)) cycle
  ldim = 2*Hubbard_l(nt) + 1

  do na = 1, nat
    if (ityp(na) /= nt) cycle
    off = offsetU(na)

    do m2 = 1, ldim
      do m1 = 1, ldim
        do ibnd = 1, nbnd
          hub_product = conjg(projU(off+m1, ibnd, ii)) * &
                        projU(off+m2, ibnd, jj)
          tmp = tmp + wg(ibnd,ik) * &
                    v%ns(m1, m2, current_spin, na) * &
                    hub_product
        enddo
      enddo
    enddo
  enddo

  delta_M_hub(kk) = delta_M_hub(kk) - 2.d0*imag(tmp)
END SUBROUTINE calc_delta_M_hub

```

For fixed Cartesian indices $\mu \equiv ii$ and $\nu \equiv jj$, this routine evaluates the quantity

$$\sum_{n\mathbf{k}}^{\text{occ}} w_{n\mathbf{k}} \sum_I \sum_{mm'} U_{mm'}^{I,\sigma} \left[\langle \partial_{k_\mu} \phi_{Im}^{\mathbf{k}} | u_{n\mathbf{k}} \rangle \right]^* \langle \partial_{k_\nu} \phi_{Im'}^{\mathbf{k}} | u_{n\mathbf{k}} \rangle,$$

where the Hubbard potential matrix $U_{mm'}^{I,\sigma}$ is stored as `v%ns(m1,m2,current_spin,na)`, while the derivative matrix elements are provided by `projU`. The corresponding contribution to the k -th component of the magnetization is then obtained as

$$\Delta M_{\text{Hub},k} \propto -2 \text{Im}[\dots].$$

6.1 Numerical test with LiCoO₂.

As a numerical benchmark, we consider the NMR shielding tensor of LiCoO₂ at the DFT+ U level, comparing QE-CONVERSE with the linear-response GIPAW implementation in CASTEP. Norm-conserving pseudopotentials and a plane-wave cutoff of 60 Ry were used, with a Hubbard correction of $U = 8$ eV applied to the Co-3d manifold. The comparison with CASTEP is particularly relevant, since its GIPAW implementation explicitly includes the non-local Hubbard contribution in the magnetic-response operator.

Table 1 reports the isotropic shielding values for the three equivalent Li, Co, and O atoms in the unit cell. For each atomic species, both methods give internally consistent values across the three symmetry-equivalent sites. The agreement is satisfactory for Li and Co, whereas a sizeable discrepancy is observed for the oxygen shielding. This is consistent with the fact that, although the Hubbard correction is applied only to the Co-3d manifold, the oxygen shielding is sensitive to the Co-3d/O-2p hybridization and therefore to the detailed definition of the Hubbard subspace. A possible contribution to the remaining discrepancy is the different construction of the Hubbard projectors and of the associated non-local magnetic-response term in the two implementations.

Table 1: ^7Li , ^{59}Co , and ^{17}O isotropic NMR shielding (ppm) in LiCoO_2 at the DFT+ U level ($U = 8$ eV on Co- $3d$), computed with QE-CONVERSE and CASTEP. Three symmetry-equivalent atoms of each species are reported.

Atom	Site	QE-CONVERSE (ppm)	CASTEP (ppm)
Li	1	92.51	87.06
Li	2	92.33	87.06
Li	3	92.32	87.06
Co	1	-4679.62	-4807.13
Co	2	-4679.64	-4807.12
Co	3	-4679.64	-4807.11
O	1	343.12	230.66
O	2	343.11	230.68
O	3	343.03	230.67

Table 2: Full NMR shielding tensor components and isotropic shieldings (ppm) for all atoms in LiCoO_2 at the DFT+ $U+V$ level, computed with QE-CONVERSE. $\sigma_{\text{iso}} = (\sigma_{xx} + \sigma_{yy} + \sigma_{zz})/3$.

Atom	σ_{xx}	σ_{yy}	σ_{zz}	σ_{iso}
Co 1	-4982.31	-4982.89	-5367.68	-5110.96
Co 2	-4982.80	-4983.56	-5367.89	-5111.41
Co 3	-4982.26	-4982.93	-5367.91	-5111.03
O 4	66.37	66.26	65.90	66.18
O 5	66.69	66.42	66.15	66.42
O 6	66.37	66.30	66.20	66.29
O 7	66.33	66.32	66.19	66.28
O 8	66.65	66.49	66.06	66.40
O 9	66.39	66.14	65.94	66.16
Li 10	88.41	88.27	82.43	86.37
Li 11	88.27	88.23	82.61	86.37
Li 12	88.24	88.24	82.65	86.38

7 DFT+U+V

In DFT+U+V, a corrective Hubbard energy E_{U+V} is added to the standard DFT energy E_{DFT} :

$$E_{\text{DFT}+U+V} = E_{\text{DFT}} + E_{U+V}. \quad (43)$$

Unlike DFT+U, which only accounts for on-site interactions scaled by U , DFT+U+V also includes inter-site interactions between an atom and its neighbors, scaled by V . In the simplified rotationally-invariant formulation, the extended Hubbard correction reads:

$$\begin{aligned} E_{U+V} = & \frac{1}{2} \sum_I \sum_{\sigma mm'} U^I \left(\delta_{mm'} - n_{mm'}^{II\sigma} \right) n_{m'm}^{II\sigma} \\ & - \frac{1}{2} \sum_I \sum_{J(J \neq I)}^* \sum_{\sigma mm'} V^{IJ} n_{mm'}^{IJ\sigma} n_{m'm}^{JI\sigma}, \end{aligned} \quad (44)$$

where I and J index atomic sites, m and m' denote magnetic quantum numbers, and U^I and V^{IJ} are the *effective* on-site and inter-site Hubbard parameters, respectively (here and everywhere in the following, U^I denotes the difference between the screened Coulomb interaction parameter and the Hund's exchange interaction parameter). The asterisk denotes that, for each atom I , the sum over J includes all neighbors up to a defined cutoff distance. The generalized occupation matrices $n_{mm'}^{IJ\sigma}$ are obtained by projecting KS states $\psi_{v,\mathbf{k}}^\sigma$ onto localized orbitals $\phi_m^I(\mathbf{r})$:

$$n_{mm'}^{IJ\sigma} = \sum_{v,\mathbf{k}} f_{v,\mathbf{k}}^\sigma \langle \psi_{v,\mathbf{k}}^\sigma | \phi_{m'}^J \rangle \langle \phi_m^I | \psi_{v,\mathbf{k}}^\sigma \rangle, \quad (45)$$

where v and σ label bands and spin, $\mathbf{k}$ runs over the first Brillouin zone (BZ), and $f_{v,\mathbf{k}}^\sigma$ are the KS occupations. In Eq. (45), $m \in I$ and $m' \in J$, meaning that m runs over the magnetic quantum numbers associated with a fixed principal quantum number n and orbital angular momentum l of atom I , and analogously m' for atom J . The corresponding generalized KS equation can be written as,

$$\left(\hat{\mathcal{H}}_{\text{KS}}^\sigma + \hat{\mathcal{V}}_{U+V}^\sigma \right) |\psi_{v,\mathbf{k}}^\sigma\rangle = \varepsilon_{v,\mathbf{k}}^\sigma |\psi_{v,\mathbf{k}}^\sigma\rangle, \quad (46)$$

where $\hat{\mathcal{H}}_{\text{KS}}^\sigma$ is the KS Hamiltonian of the standard DFT, $\varepsilon_{v,\mathbf{k}}^\sigma$ is the v -th band energy at momentum $\mathbf{k}$ and spin σ , and $\hat{\mathcal{V}}_{U+V}^\sigma$ is the Hubbard potential that is derived from the Hubbard energy (44) and which reads:

$$\begin{aligned} \hat{\mathcal{V}}_{U+V}^\sigma = & \sum_I \sum_{mm'} \frac{U^I}{2} \left(\delta_{mm'} - 2n_{mm'}^{II\sigma} \right) |\phi_m^I\rangle \langle \phi_{m'}^I| \\ & - \sum_I \sum_{J(J \neq I)}^* \sum_{mm'} V^{IJ} n_{mm'}^{IJ\sigma} |\phi_m^I\rangle \langle \phi_{m'}^J|. \end{aligned} \quad (47)$$

In Eqs. (44) and (47), the effective on-site U^I and inter-site V^{IJ} terms act in opposition when both parameters are positive: the on-site term favors electron localization on atomic sites and suppresses hybridization, whereas the inter-site term enhances hybridization with neighboring atoms. Accurate first-principles evaluation of U^I and V^{IJ} is therefore crucial to correctly describe the balance between localization and hybridization.

It is important to note that the computed values of U^I and V^{IJ} depend on the choice of Hubbard projector functions $\phi_m^I(\mathbf{r})$ used to define the occupation matrix in Eq. (45) and the Hubbard potential in Eq. (47).

7.1 Contribution to the orbital magnetization

We now derive the contribution of the full DFT+U + V potential to the orbital magnetization, generalizing the DFT+U result of Section 6.

Notation. It is convenient to introduce a unified potential matrix that covers both the on-site ($J = I$) and inter-site ($J \neq I$) terms of Eq. (47):

$$W_{mm'}^{IJ,\sigma} = \begin{cases} \frac{U^I}{2} (\delta_{mm'} - 2n_{mm'}^{II\sigma}) & J = I, \\ -V^{IJ} n_{mm'}^{IJ\sigma} & J \neq I. \end{cases} \quad (48)$$

In QE the full matrix $W_{mm'}^{IJ,\sigma}$ is stored as `v_nsg(m,m',viz,na,σ)`, where `na` = I and `viz` is the neighbor index such that `neighbor(na)%neigh(viz) = J`. The on-site entry corresponds to the self-neighbor (`neigh(viz) = I`), so the sum over all viz automatically includes both parts.

Using $W_{mm'}^{IJ}$, the DFT+ U + V potential reads

$$\hat{\mathcal{V}}_{U+V}^\sigma = \sum_I \sum_J^* \sum_{mm'} W_{mm'}^{IJ,\sigma} |\phi_m^I\rangle \langle \phi_{m'}^J|, \quad (49)$$

where the asterisk on $\sum_J$ again denotes the sum over the self-site ($J = I$) and all neighbors within the cutoff.

On-site part ($J = I$). The on-site contribution has the same operator structure as the DFT+ U potential: both bra and ket belong to atom I . The derivation of Section 6 therefore applies unchanged, with $V_{mm'}^I \rightarrow W_{mm'}^{II,\sigma}$. It yields Eq. (42) with this replacement.

Inter-site part ($J \neq I$). For the inter-site terms, the ket $|\phi_m^I\rangle$ belongs to atom I but the bra $\langle \phi_{m'}^J|$ belongs to atom $J \neq I$. Starting from Eq. (13), the inter-site contribution to the component γ of the orbital magnetization reads

$$\Delta M_{V,\gamma} = -\frac{\alpha}{2} \sum_{n\mathbf{k}}^{\text{occ}} f_{n\mathbf{k}} \sum_I \sum_{J \neq I}^* \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \frac{1}{i} \langle u_{n\mathbf{k}} | \rho_{I,\mu} | \tilde{\phi}_m^{\mathbf{k},I} \rangle W_{mm'}^{IJ,\sigma} \langle \tilde{\phi}_{m'}^{\mathbf{k},J} | \rho_{I,\nu} | u_{n\mathbf{k}} \rangle. \quad (50)$$

The left matrix element is handled exactly as in the DFT+ U case via the no-phase identity (39):

$$\langle u_{n\mathbf{k}} | \rho_{I,\mu} | \tilde{\phi}_m^{\mathbf{k},I} \rangle = i \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle. \quad (51)$$

For the right matrix element we decompose the relative coordinate with respect to I into the relative coordinate with respect to J plus the inter-site displacement $\mathbf{d}^{IJ} = \mathbf{R}_J - \mathbf{R}_I$:

$$\rho_{I,\nu} = \rho_{J,\nu} + d_\nu^{IJ}, \quad \mathbf{d}^{IJ} = \mathbf{R}_J - \mathbf{R}_I. \quad (52)$$

Applying the no-phase identity for atom J , Eq. (40), to the first term, and collecting:

$$\langle \tilde{\phi}_{m'}^{\mathbf{k},J} | \rho_{I,\nu} | u_{n\mathbf{k}} \rangle = -i \langle \partial_{k_\nu} \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle + d_\nu^{IJ} \langle \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle. \quad (53)$$

The second term is new: it involves the regular (non-derivative) Hubbard overlap

$$s_{m',n}^J(\mathbf{k}) \equiv \langle \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle, \quad (54)$$

which is the no-phase analogue of the standard $\langle \beta | u \rangle$ projector overlap.

Substituting Eqs. (51)–(53) into Eq. (50):

$$\Delta M_{V,\gamma} = -\frac{\alpha}{2} \sum_{n\mathbf{k}}^{\text{occ}} f_{n\mathbf{k}} \sum_{I,J \neq I}^* \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle W_{mm'}^{IJ,\sigma} \left[-i \langle \partial_{k_\nu} \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle + d_\nu^{IJ} s_{m',n}^J \right]. \quad (55)$$

This splits into two structurally distinct contributions.

Part A: derivative–derivative term. The first bracket term gives

$$\Delta M_{V,\gamma}^A = \frac{i\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_{I,J \neq I}^* \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle W_{mm'}^{IJ,\sigma} \langle \partial_{k_\nu} \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle, \quad (56)$$

which has exactly the same structure as the DFT+ U result Eq. (41), now with projector derivatives from two *different* atomic centers. Passing to the imaginary part as in Section 6:

$$\Delta M_{V,\gamma}^A = -\frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_{I,J \neq I}^* \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \text{Im} \left[\langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle W_{mm'}^{IJ,\sigma} \langle \partial_{k_\nu} \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle \right]. \quad (57)$$

Part B: derivative–overlap term. The second bracket term gives

$$\Delta M_{V,\gamma}^B = -\frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_{I,J \neq I}^* \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} d_\nu^{IJ} \langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle W_{mm'}^{IJ,\sigma} s_{m',n}^J. \quad (58)$$

This term is absent in the pure DFT+ U case (where $J = I$ so $\mathbf{d}^{II} = \mathbf{0}$) and arises solely from the inter-site nature of the projector pair. Since d_ν^{IJ} is real while $W_{mm'}^{IJ,\sigma}$ and $s_{m',n}^J$ are in general complex, the expression under the sum is complex; the physical result is its real part (the imaginary parts cancel in the full sum over ordered pairs (I, J) and (J, I) , as follows from $W_{m'm}^{JI} = (W_{mm'}^{IJ})^*$ and time-reversal):

$$\Delta M_{V,\gamma}^B = -\frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_{I,J \neq I}^* \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} d_\nu^{IJ} \text{Re} \left[\langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle W_{mm'}^{IJ,\sigma} s_{m',n}^J \right]. \quad (59)$$

Full DFT+ U + V magnetization formula. Combining the on-site part (Eq. (42) with $V^I \rightarrow W^{II}$) with Parts A and B, the total contribution reads

$$\begin{aligned} \Delta M_{U+V,\gamma} = & -\frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_I^* \sum_J \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} \text{Im} \left[\langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle W_{mm'}^{IJ,\sigma} \langle \partial_{k_\nu} \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle \right] \\ & - \frac{\alpha}{2} \sum_{n\mathbf{k}} f_{n\mathbf{k}} \sum_I^* \sum_{J \neq I} \sum_{mm'} \sum_{\mu\nu} \epsilon_{\gamma\mu\nu} d_\nu^{IJ} \text{Re} \left[\langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle W_{mm'}^{IJ,\sigma} s_{m',n}^J \right]. \end{aligned} \quad (60)$$

The first line (the Im term) extends over *all* neighbors including $J = I$; for $J = I$ it reduces to Eq. (42) since $W^{II} = V^{II}$ (the DFT+ U potential matrix). The second line (the Re term) is strictly inter-site and vanishes when $V = 0$.

Implementation notes.

- The derivative overlaps $\langle u_{n\mathbf{k}} | \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I} \rangle$ and $\langle \partial_{k_\nu} \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle$ are already computed by `compute_dhubbecp` for every Hubbard atom.
- The regular overlap $s_{m',n}^J = \langle \tilde{\phi}_{m'}^{\mathbf{k},J} | u_{n\mathbf{k}} \rangle$ must be computed separately, e.g. by calling `orthoUwfc_k` at the unshifted $\mathbf{k}$, stripping the phase via `strip_kphase_wfcU`, and projecting with `calbec`.
- The potential matrix $W_{mm'}^{IJ,\sigma}$ is stored as `v_nsg(m,m',viz,na,σ)`, where `na` = I and `neighborhood(na)%neighborhood(J)`.
- The inter-site displacement $\mathbf{d}^{IJ} = \mathbf{R}_J - \mathbf{R}_I$ is available from `tau(:,na2) - tau(:,na1)` in crystal coordinates (converted to Cartesian with `alat`).

Appendix

A Phase convention for the k -derivative of Kleinman–Bylander projectors

In this Appendix we discuss the phase convention used to compute the k -derivative of Kleinman–Bylander (KB) projectors. The purpose is to show that the derivative entering the GIPAW/converse expression must be taken with respect to the local coordinate measured from the atomic center. This requires removing the global phase associated with the position of the atomic center.

Let $\mathbf{r}$ denote the electronic coordinate and let $\mathbf{R}_I$ be the position of atom I . We define the local coordinate with respect to atom I as

$$\boldsymbol{\rho}_I = \mathbf{r} - \mathbf{R}_I. \quad (61)$$

The localized projector centered on atom I is denoted by

$$\beta_i^I(\mathbf{r}) \equiv \beta_i(\mathbf{r} - \mathbf{R}_I) = \beta_i(\boldsymbol{\rho}_I). \quad (62)$$

The GIPAW contribution involving a non-local operator centered on atom I contains the relative position operator

$$\mathbf{r} - \mathbf{R}_I = \boldsymbol{\rho}_I, \quad (63)$$

rather than the absolute position $\mathbf{r}$. Therefore, the k -derivative of the projector must reproduce the action of $\boldsymbol{\rho}_I$.

A.1 Full-phase projector

Consider first the projector with the full Bloch phase. In real space, this object may be written as

$$\beta_{i,\text{full}}^{\mathbf{k},I}(\mathbf{r}) = e^{-i\mathbf{k}\cdot\mathbf{r}} \beta_i^I(\mathbf{r}). \quad (64)$$

Using Eq. (62), this can also be written as

$$\beta_{i,\text{full}}^{\mathbf{k},I}(\mathbf{r}) = e^{-i\mathbf{k}\cdot\mathbf{r}} \beta_i(\mathbf{r} - \mathbf{R}_I). \quad (65)$$

Taking the derivative with respect to k_μ gives

$$\partial_{k_\mu} \beta_{i,\text{full}}^{\mathbf{k},I}(\mathbf{r}) = -i r_\mu \beta_{i,\text{full}}^{\mathbf{k},I}(\mathbf{r}). \quad (66)$$

Thus, the full-phase derivative generates the absolute coordinate r_μ , not the relative coordinate $r_\mu - R_{I,\mu}$.

The same result can be seen in reciprocal space. The plane-wave coefficients of the full-phase projector have the structure

$$\beta_{i,\text{full}}^{\mathbf{k},I}(\mathbf{G}) = e^{-i(\mathbf{k}+\mathbf{G})\cdot\mathbf{R}_I} \beta_i(\mathbf{k} + \mathbf{G}), \quad (67)$$

where $\beta_i(\mathbf{k} + \mathbf{G})$ is the Fourier transform of the projector centered at the origin. Separating the phase factor gives

$$\beta_{i,\text{full}}^{\mathbf{k},I}(\mathbf{G}) = e^{-i\mathbf{k}\cdot\mathbf{R}_I} e^{-i\mathbf{G}\cdot\mathbf{R}_I} \beta_i(\mathbf{k} + \mathbf{G}). \quad (68)$$

Taking the derivative with respect to k_μ yields

$$\begin{aligned} \partial_{k_\mu} \beta_{i,\text{full}}^{\mathbf{k},I}(\mathbf{G}) &= \partial_{k_\mu} \left[e^{-i\mathbf{k}\cdot\mathbf{R}_I} e^{-i\mathbf{G}\cdot\mathbf{R}_I} \beta_i(\mathbf{k} + \mathbf{G}) \right] \\ &= e^{-i\mathbf{k}\cdot\mathbf{R}_I} e^{-i\mathbf{G}\cdot\mathbf{R}_I} \partial_{k_\mu} \beta_i(\mathbf{k} + \mathbf{G}) \\ &\quad - i R_{I,\mu} e^{-i\mathbf{k}\cdot\mathbf{R}_I} e^{-i\mathbf{G}\cdot\mathbf{R}_I} \beta_i(\mathbf{k} + \mathbf{G}). \end{aligned} \quad (69)$$

The second term in Eq. (69) is explicitly proportional to the atomic position $R_{I,\mu}$. This contribution originates from the derivative of the global phase

$$e^{-i\mathbf{k}\cdot\mathbf{R}_I}. \quad (70)$$

It is not associated with the intrinsic spatial dependence of the localized projector, but only with the absolute position of its center.

This term is undesirable for the GIPAW/converse expression, because the operator entering the commutator is $\mathbf{r} - \mathbf{R}_I$, not $\mathbf{r}$. Equivalently, the derivative must be insensitive to an arbitrary global phase attached to the atomic center. Keeping the full phase would introduce an origin-dependent contribution proportional to $\mathbf{R}_I$.

A.2 No-phase projector

To remove this spurious contribution, one defines the no-phase projector by dropping the factor $e^{-i\mathbf{k}\cdot\mathbf{R}_I}$ from Eq. (68). In reciprocal space, the no-phase projector is therefore

$$\tilde{\beta}_i^{\mathbf{k},I}(\mathbf{G}) = e^{-i\mathbf{G}\cdot\mathbf{R}_I} \beta_i(\mathbf{k} + \mathbf{G}). \quad (71)$$

This is the object constructed by the no-phase version of the KB projector initialization. In the code, the factor $e^{-i\mathbf{G}\cdot\mathbf{R}_I}$ is the structure factor associated with the reciprocal lattice vector $\mathbf{G}$, whereas the phase $e^{-i\mathbf{k}\cdot\mathbf{R}_I}$ is deliberately omitted.

Taking the derivative of Eq. (71) gives

$$\partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{G}) = e^{-i\mathbf{G}\cdot\mathbf{R}_I} \partial_{k_\mu} \beta_i(\mathbf{k} + \mathbf{G}). \quad (72)$$

Unlike Eq. (69), there is no term proportional to $R_{I,\mu}$. Thus, the derivative acts only on the intrinsic dependence of the projector on $\mathbf{k} + \mathbf{G}$.

A.3 Real-space reconstruction of the no-phase projector

We now show explicitly that the no-phase reciprocal-space definition corresponds to a real-space projector whose k -derivative generates the relative coordinate $\mathbf{r} - \mathbf{R}_I$.

The inverse plane-wave expansion of Eq. (71) is

$$\tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{G}} \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}}, \quad (73)$$

where Ω is the cell volume. Substituting Eq. (71), one obtains

$$\begin{aligned} \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) &= \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{G}} e^{-i\mathbf{G}\cdot\mathbf{R}_I} \beta_i(\mathbf{k} + \mathbf{G}) e^{i\mathbf{G}\cdot\mathbf{r}} \\ &= \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{G}} \beta_i(\mathbf{k} + \mathbf{G}) e^{i\mathbf{G}\cdot(\mathbf{r}-\mathbf{R}_I)}. \end{aligned} \quad (74)$$

Using the local coordinate $\boldsymbol{\rho}_I = \mathbf{r} - \mathbf{R}_I$, this becomes

$$\tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) = \frac{1}{\sqrt{\Omega}} \sum_{\mathbf{G}} \beta_i(\mathbf{k} + \mathbf{G}) e^{i\mathbf{G}\cdot\boldsymbol{\rho}_I}. \quad (75)$$

The quantity $\beta_i(\mathbf{k} + \mathbf{G})$ is the Fourier coefficient of the localized function multiplied by the Bloch phase relative to the atomic center. Hence Eq. (75) is the plane-wave representation of

$$\tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) = e^{-i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R}_I)} \beta_i(\mathbf{r} - \mathbf{R}_I), \quad (76)$$

up to the usual periodic repetition implied by the plane-wave representation.

Taking the derivative of Eq. (76) gives

$$\begin{aligned}
\partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) &= \partial_{k_\mu} \left[e^{-i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R}_I)} \beta_i(\mathbf{r}-\mathbf{R}_I) \right] \\
&= -i(r_\mu - R_{I,\mu}) e^{-i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R}_I)} \beta_i(\mathbf{r}-\mathbf{R}_I) \\
&= -i(r_\mu - R_{I,\mu}) \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) \\
&= -i\rho_{I,\mu} \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}).
\end{aligned} \tag{77}$$

Therefore,

$$(r_\mu - R_{I,\mu}) \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}) = i \partial_{k_\mu} \tilde{\beta}_i^{\mathbf{k},I}(\mathbf{r}). \tag{78}$$

Equation (78) is the identity required in the GIPAW/converse derivation. It shows that the no-phase derivative generates the relative position operator with respect to the atomic center.

A.4 No-phase convention for Hubbard projectors

The same phase convention must be used for the Hubbard projectors. Let ϕ_{Im} be a localized Hubbard orbital centered on atom I . In reciprocal space, the full-phase Hubbard projector has the same translation structure as a KB projector:

$$\phi_{m,\text{full}}^{\mathbf{k},I}(\mathbf{G}) = e^{-i(\mathbf{k}+\mathbf{G})\cdot\mathbf{R}_I} \phi_m(\mathbf{k} + \mathbf{G}). \tag{79}$$

Separating the phase associated with the atomic center gives

$$\phi_{m,\text{full}}^{\mathbf{k},I}(\mathbf{G}) = e^{-i\mathbf{k}\cdot\mathbf{R}_I} e^{-i\mathbf{G}\cdot\mathbf{R}_I} \phi_m(\mathbf{k} + \mathbf{G}). \tag{80}$$

The factor $e^{-i\mathbf{G}\cdot\mathbf{R}_I}$ must be retained, because it places the localized orbital on atom I . By contrast, the factor $e^{-i\mathbf{k}\cdot\mathbf{R}_I}$ is a global, k -dependent phase associated with the position of the atomic center. If differentiated with respect to $\mathbf{k}$, it generates an additional term proportional to $\mathbf{R}_I$.

Therefore we define the no-phase Hubbard projector as

$$\tilde{\phi}_m^{\mathbf{k},I}(\mathbf{G}) = e^{+i\mathbf{k}\cdot\mathbf{R}_I} \phi_{m,\text{full}}^{\mathbf{k},I}(\mathbf{G}), \tag{81}$$

or equivalently

$$\tilde{\phi}_m^{\mathbf{k},I}(\mathbf{G}) = e^{-i\mathbf{G}\cdot\mathbf{R}_I} \phi_m(\mathbf{k} + \mathbf{G}). \tag{82}$$

With this definition, the inverse plane-wave expansion gives the cell-periodic projector

$$\tilde{\phi}_m^{\mathbf{k},I}(\mathbf{r}) = e^{-i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R}_I)} \phi_m(\mathbf{r}-\mathbf{R}_I), \tag{83}$$

up to the periodic repetition implied by the plane-wave representation. Consequently,

$$\partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I}(\mathbf{r}) = -i(r_\mu - R_{I,\mu}) \tilde{\phi}_m^{\mathbf{k},I}(\mathbf{r}). \tag{84}$$

Thus,

$$(r_\mu - R_{I,\mu}) \tilde{\phi}_m^{\mathbf{k},I}(\mathbf{r}) = i \partial_{k_\mu} \tilde{\phi}_m^{\mathbf{k},I}(\mathbf{r}). \tag{85}$$

This is the same identity used for the no-phase KB projectors. Therefore, when the Hubbard contribution is written in terms of $\partial_{\mathbf{k}} \phi_{Im}^{\mathbf{k}}$, the derivative must be taken on the no-phase Hubbard projectors. Otherwise the derivative would also act on the global phase $e^{-i\mathbf{k}\cdot\mathbf{R}_I}$ and would produce a spurious contribution proportional to $R_{I,\mu}$.